



PROITBRIDGE
Strive For Better Future

The Math They Skipped When They Taught You to **CODE.**



AI Coach John
Follow for More

You can train a model in 10 lines —
until the loss won't move.
This is what sits underneath.

THE 6 PILLARS

1



LINEAR ALGEBRA
represent the data

2



CALCULUS
how models learn

3



PROBABILITY
handle uncertainty

4



STATISTICS
is the result real?

5



INFORMATION THEORY
where loss comes from

6

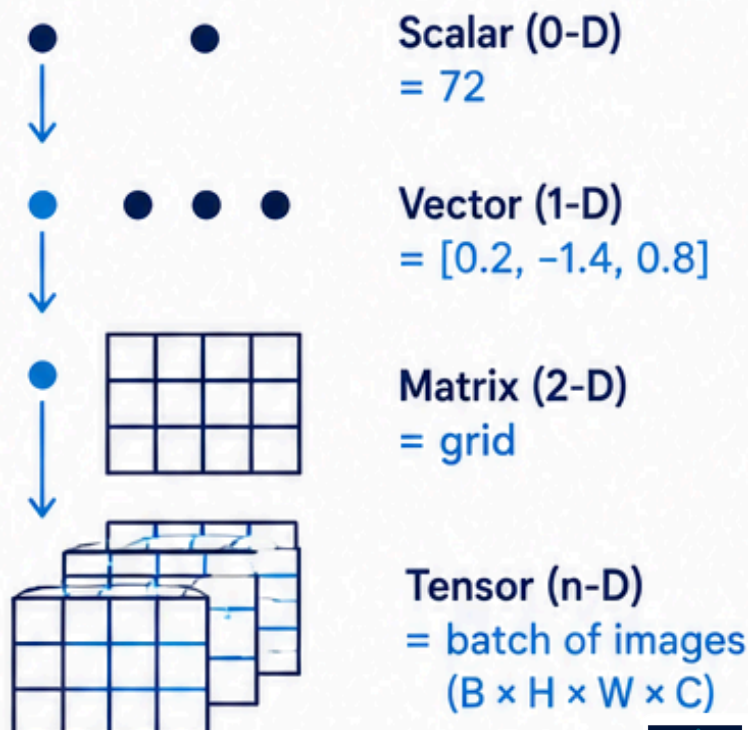


OPTIMIZATION
how models improve

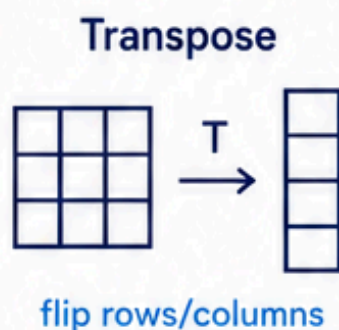
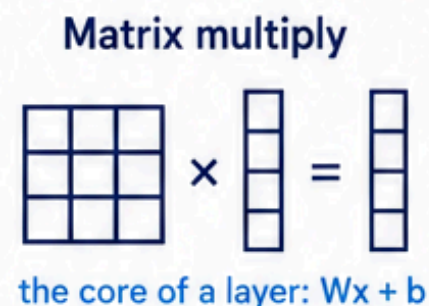
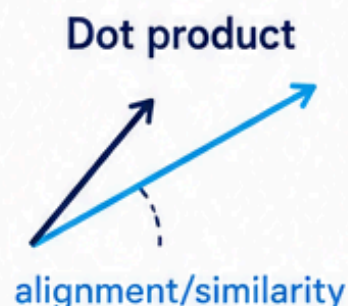
LINEAR ALGEBRA

How data becomes something a model can do math on.

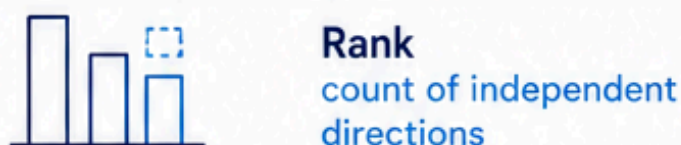
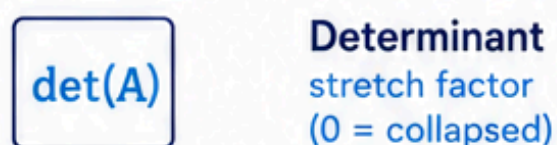
1 THE BUILDING BLOCKS



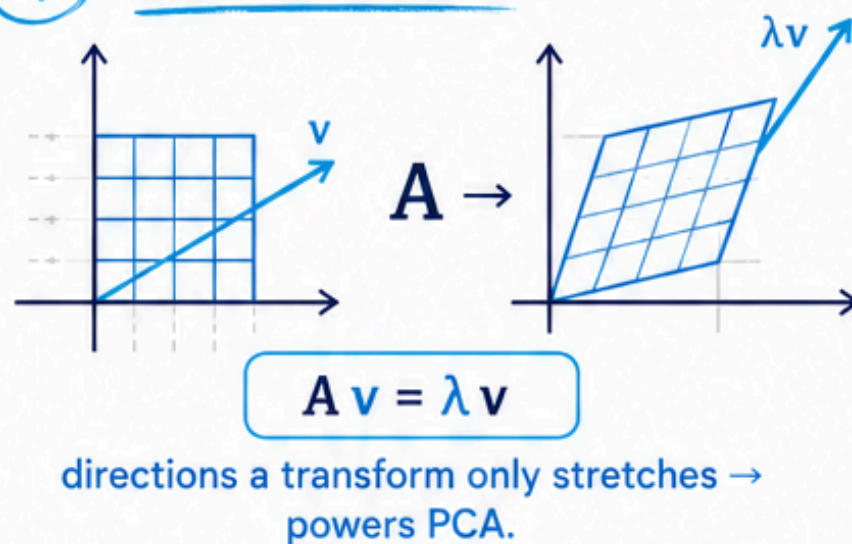
2 CORE OPERATIONS



3 CAN IT BE UNDONE?



4 EIGENVECTORS



5 SVD



keep top-k → compression;
basis of LoRA & recommenders.

6 NORMS & DISTANCE

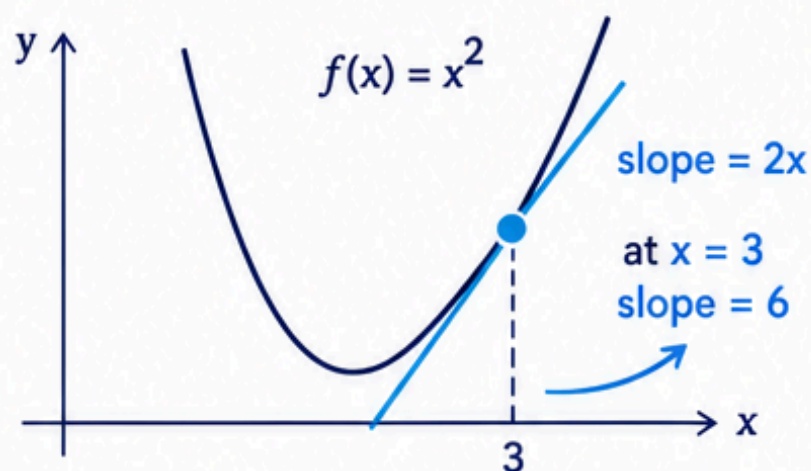


L2 powers cosine similarity &
vector search (RAG).

CALCULUS FOR ML

The slope that tells a model which way to adjust.

1 DERIVATIVES

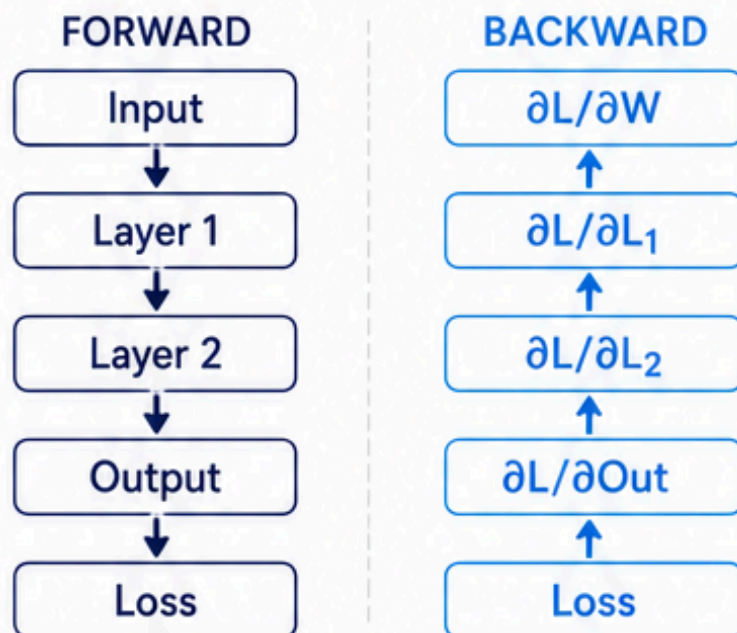


$$f(x) = x^2 \rightarrow f'(x) = 2x \rightarrow \text{at } x = 3, \text{ slope} = 6$$



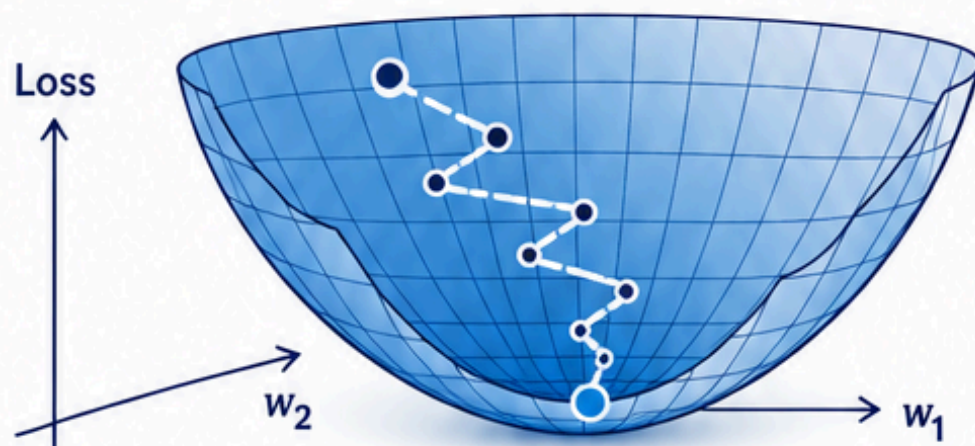
partial derivative = nudge
ONE input, hold the rest still.

2 GRADIENT & CHAIN RULE



gradient = all partials stacked;
chain rule = backpropagation.

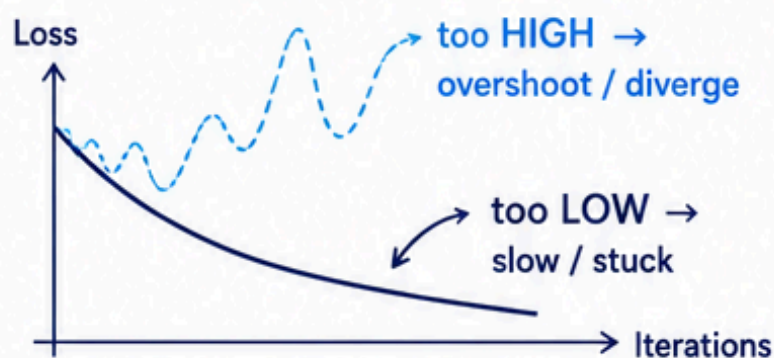
3 GRADIENT DESCENT



- 1 init
- 2 loss
- 3 gradient
- 4 $w = w - (lr \times \text{grad})$
- 5 repeat



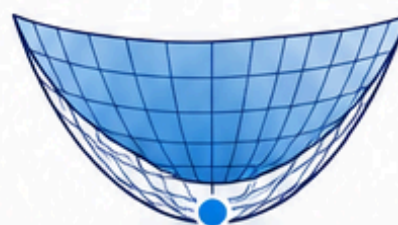
4 LEARNING RATE



most important hyperparameter

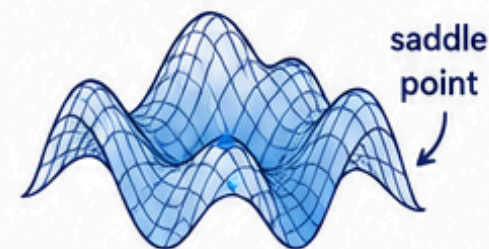
5 THE LANDSCAPE

Convex (smooth bowl)
one minimum



Hessian = curvature

Non-convex (bumpy)
many valleys

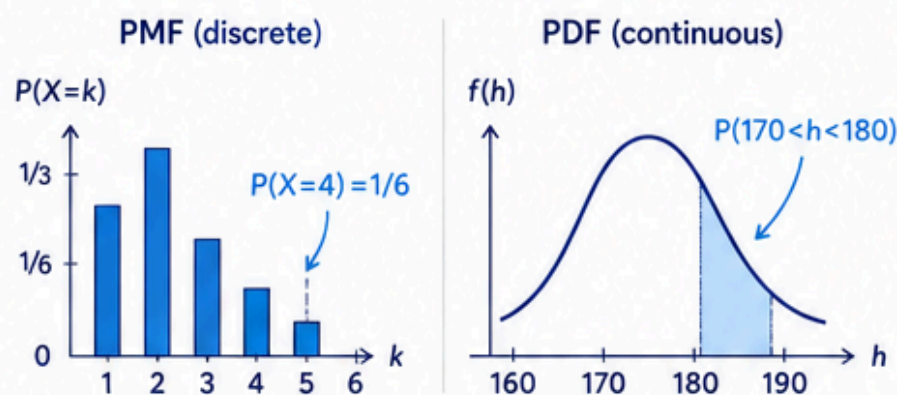


Jacobian = gradient
for multi-output

PROBABILITY

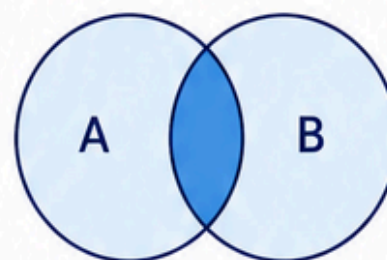
Putting numbers on uncertainty.

1 PMF vs PDF



💡 softmax = a PMF.

2 HOW EVENTS RELATE



- joint = overlap
- marginal = one circle
- conditional $P(A|B)$ = updated belief
- independent $\rightarrow P(A) \cdot P(B)$



3 BAYES' THEOREM

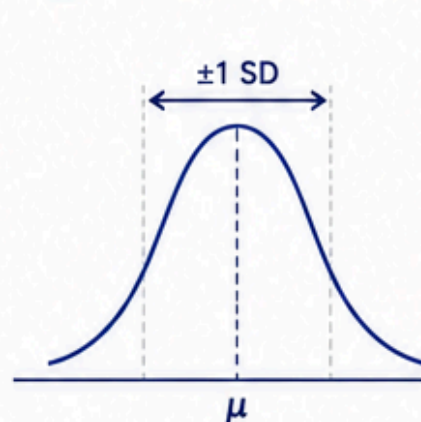
$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$



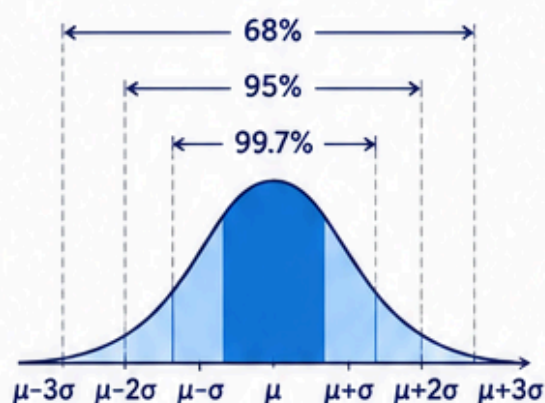
- rare disease 1%
- test 95% accurate
- positive $\rightarrow$ truly sick $\approx$ **16%**

4 DESCRIBE A DISTRIBUTION



- expectation = center
- variance = spread
- σ SD = $\sqrt{\text{variance}}$
- covariance = move together?

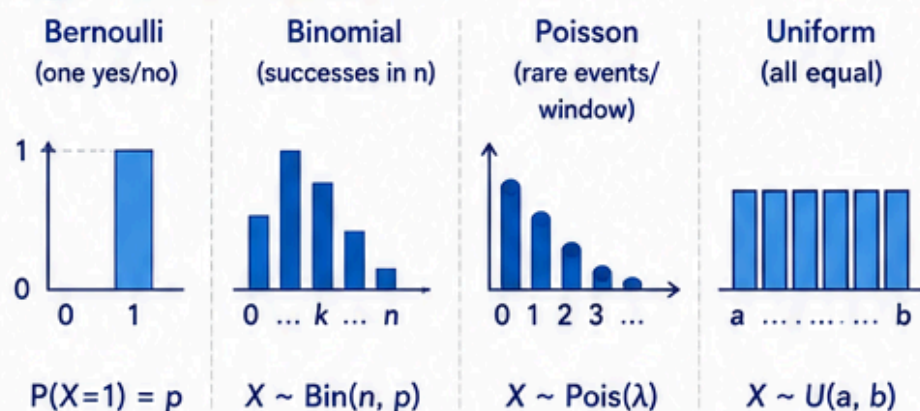
5 NORMAL DISTRIBUTION



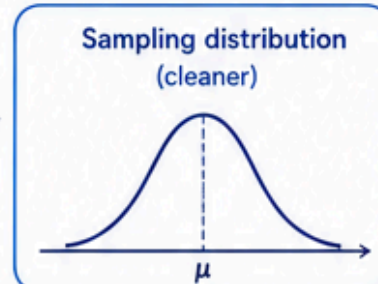
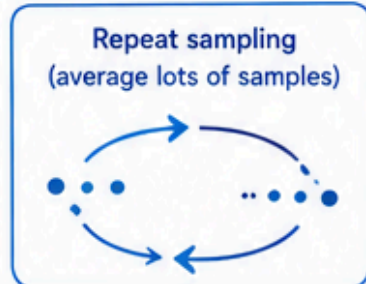
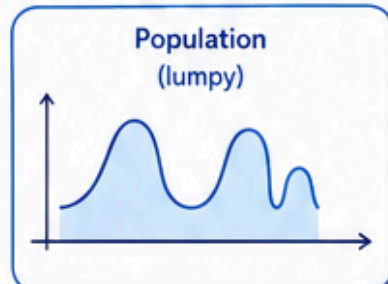
- $\pm 1 \text{ SD} \approx 68\%$
- $\pm 2 \text{ SD} \approx 95\%$
- $\pm 3 \text{ SD} \approx 99.7\%$

💡 weight init & diffusion noise.

6 4 DISTRIBUTIONS



7 CLT & LLN



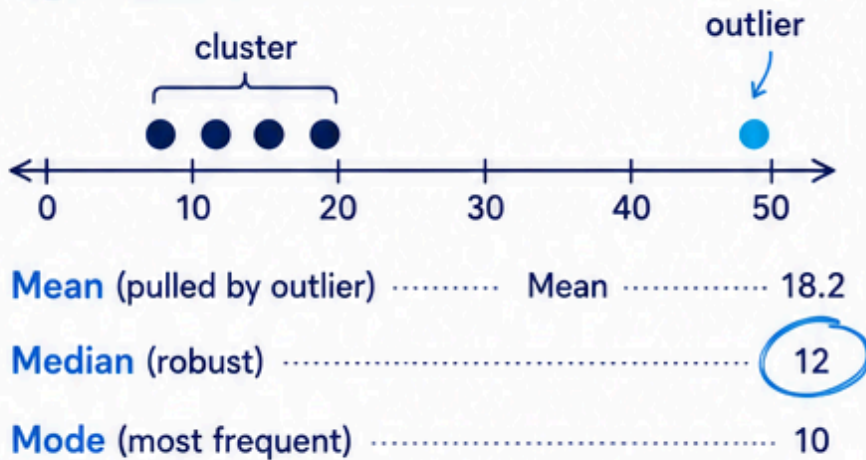
CLT =
sample means $\rightarrow$ normal

LLN =
more data $\rightarrow$
true mean

STATISTICS

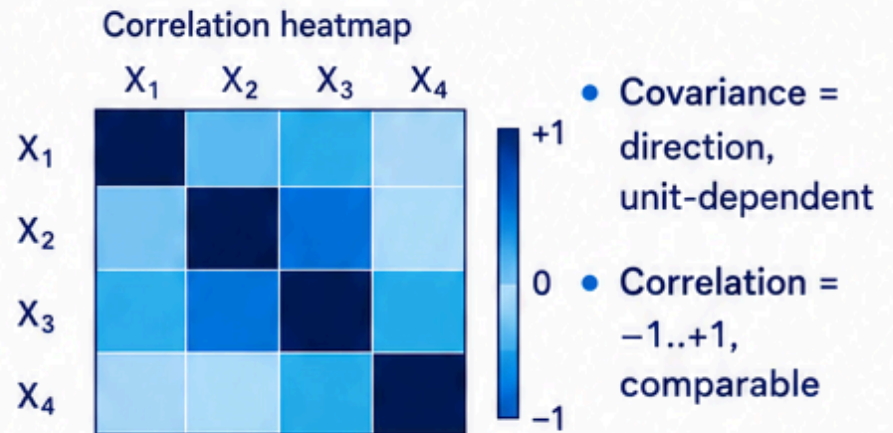
Is this improvement real, or just noise?

1 WHAT'S TYPICAL?



impute skewed features with the median.

2 CORRELATION vs COVARIANCE

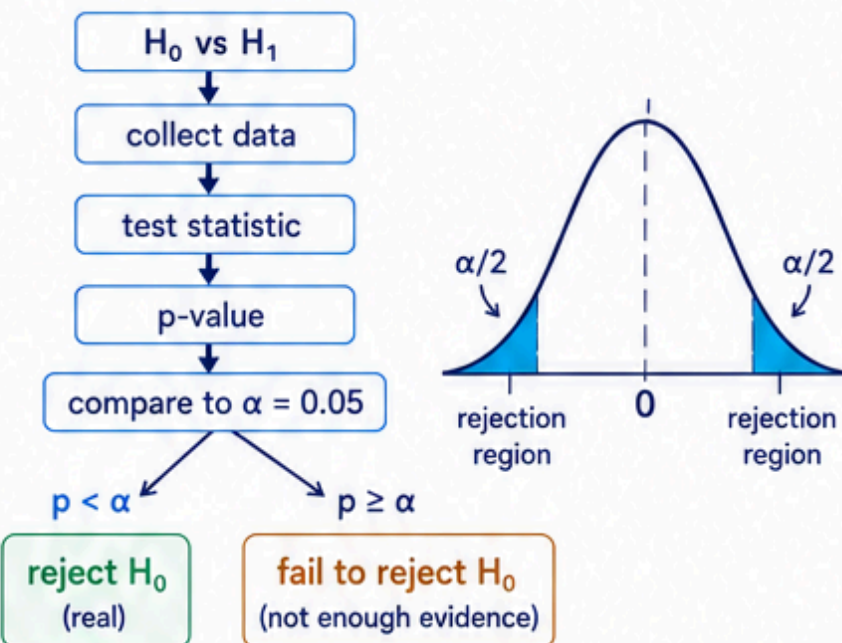


spot multicollinearity → drop a feature.



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3 HYPOTHESIS TESTING



4 READING THE NUMBERS



p-value = $P(\text{data this extreme} \mid H_0 \text{ true})$, NOT $P(H_0 \text{ true})$



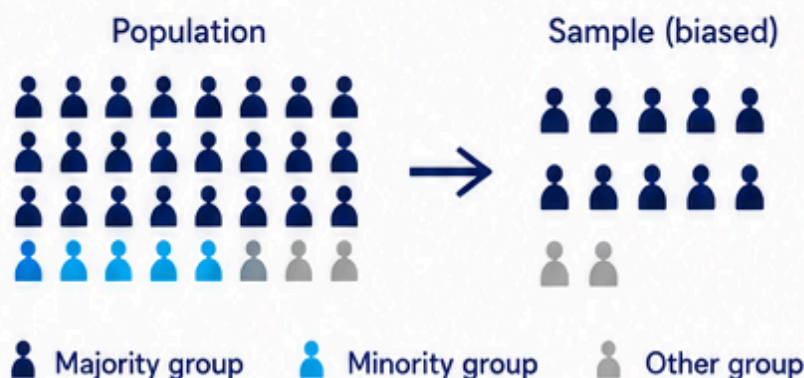
confidence interval = plausible range for the truth



z-score = how many SDs from the mean.

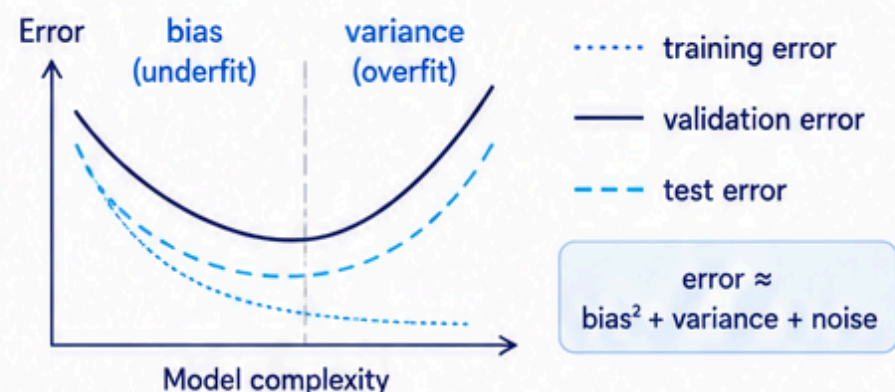
Example: **+2.3 pts** (95% CI 1.1–3.5, $p = 0.002$)

5 SAMPLING & BIAS



use stratified splits; audit before training.

6 BIAS-VARIANCE + MLE vs MAP



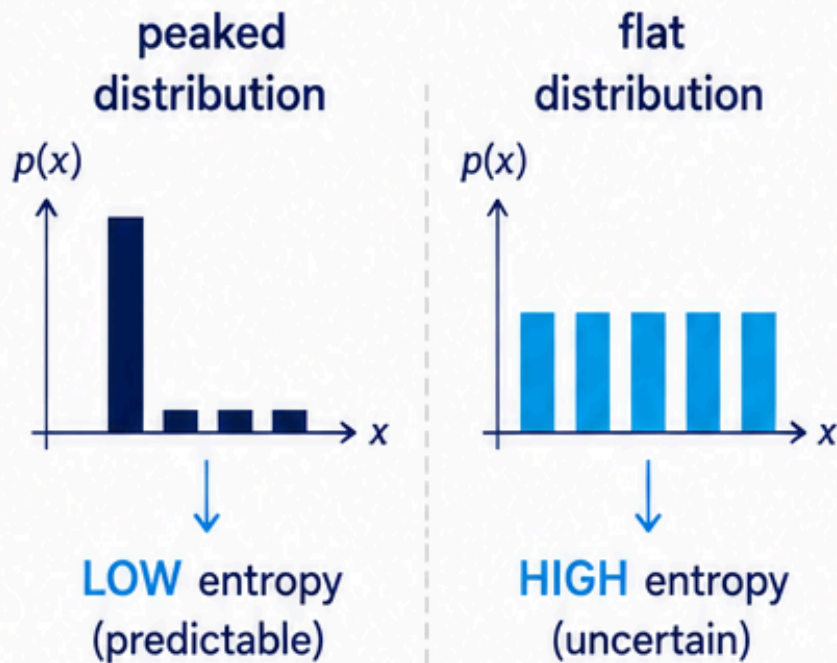
MLE = data only

MAP = data + prior
(= L2 regularization)

INFORMATION THEORY

Where loss functions actually come from.

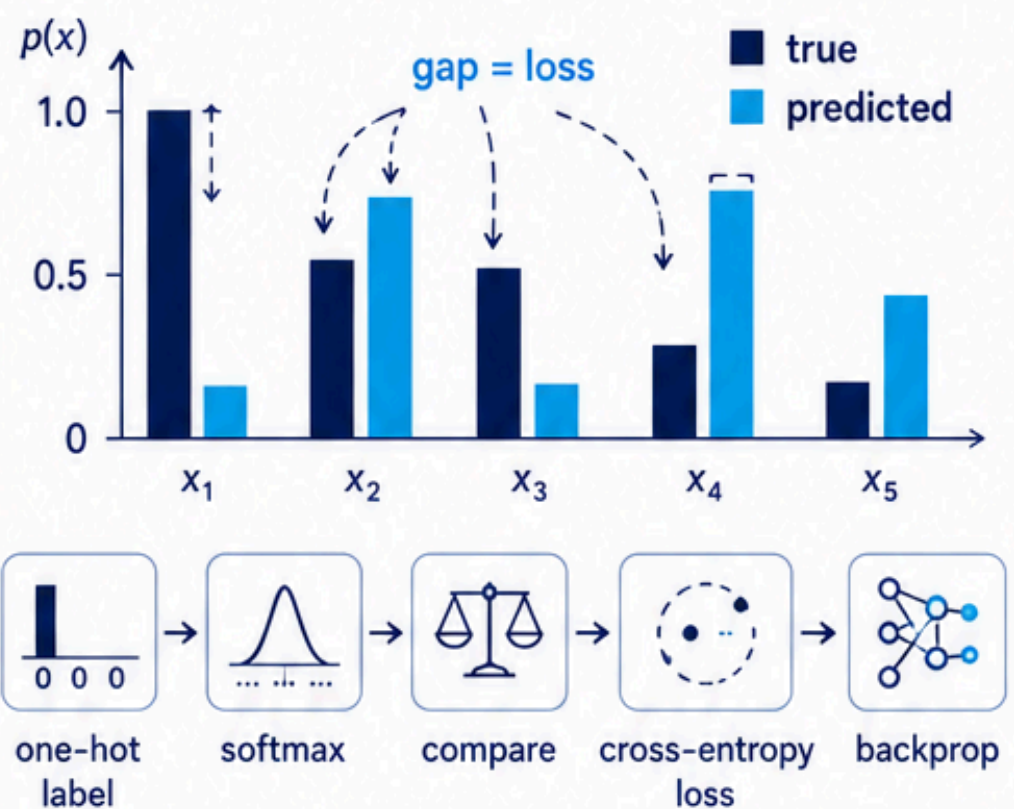
1 ENTROPY



$$H(p) = - \sum_x p(x) \log p(x)$$

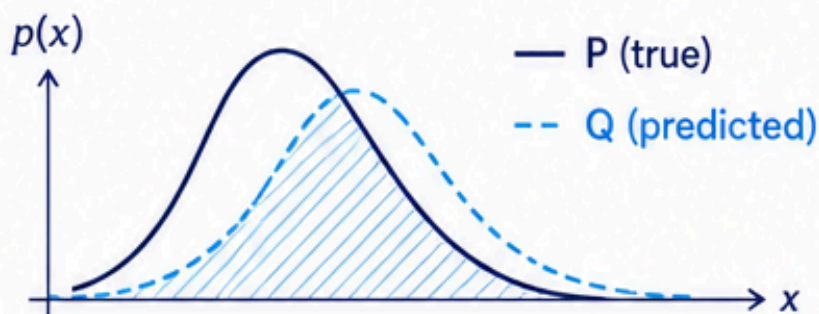


2 CROSS-ENTROPY



THE loss for classification & LLMs.

3 KL DIVERGENCE



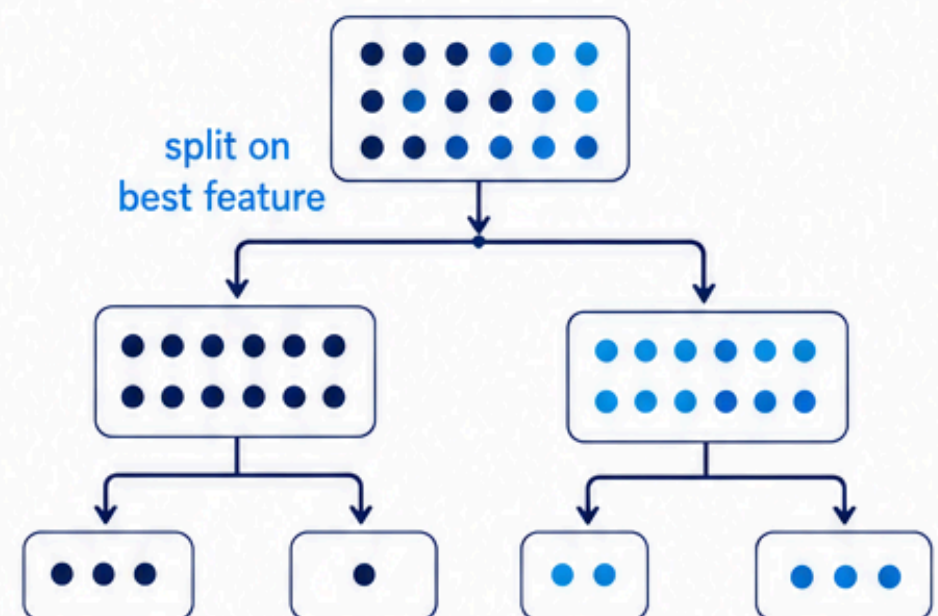
$$D_{KL}(P \parallel Q) = \sum_x P(x) \cdot \log \frac{P(x)}{Q(x)}$$

Cross-Entropy = Entropy + KL



not symmetric; used in VAEs, RLHF, distillation.

4 INFORMATION GAIN



reduction in entropy after a feature → how trees pick splits.

OPTIMIZATION

How models actually get trained in practice.

1 LOSS vs COST



Loss =
error on
ONE example



Cost =
average loss
over the batch
(what we **minimize**)



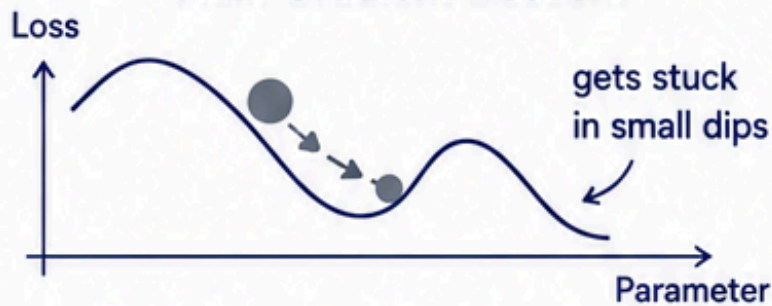
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2 GRADIENT DESCENT VARIANTS

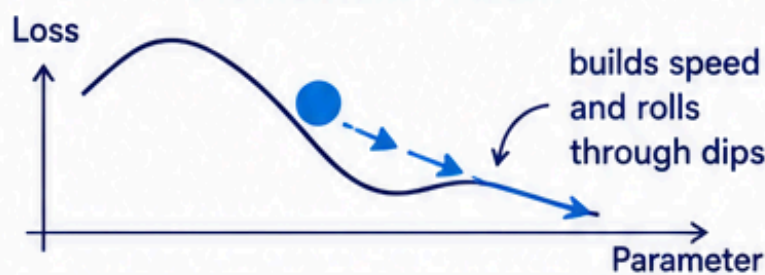
Variant	Uses	Pros / Cons
Batch 	whole dataset	stable but slow 
Stochastic (SGD) 	one example	fast but noisy 
Mini-batch (32–256) 	the practical default	balanced 

3 MOMENTUM & ADAM

Plain Gradient Descent



Momentum / Adam



Momentum =
average of past gradients

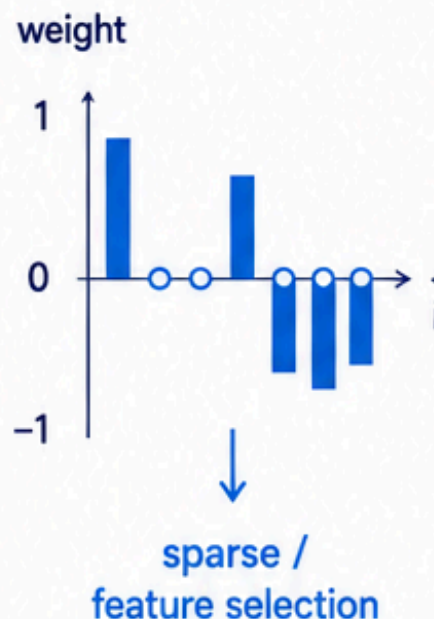


Adam =
momentum + per-parameter
adaptive LR

4 L1 vs L2 REGULARIZATION

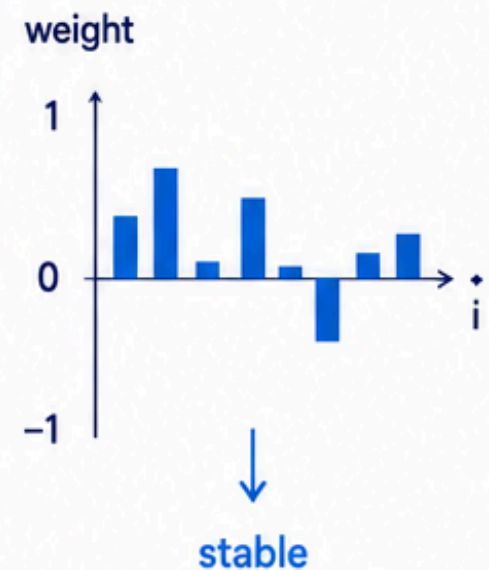
L1 (Lasso)

many weights
exactly 0



L2 (Ridge)

all weights shrink,
none exactly 0



fights overfitting;
PyTorch `weight_decay = L2`.

You Just Built the Entire MATH FOUNDATION of AI Engineering.

Linear algebra represents the data. Calculus makes it learn.

Probability & stats tell you what to trust.

Information theory is where loss comes from.

RECAP



LINEAR
ALGEBRA



CALCULUS



PROBABILITY



STATISTICS



INFO
THEORY



OPTIMIZATION



Every system you'll build —
RAG, fine-tuned LLM,
vision model —
rests on these.